Investigating Impact of Weather on Water Quality Prediction – A Pre- and Post-COVID Analysis

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***Abstract*—The Water Quality Index (WQI) is a key measure used to understand the health of water bodies and ensure safe access to water. Most existing models for predicting WQI focus only on water-based measurements, often ignoring outside environmental factors like weather. This study looks at how adding weather data—temperature, humidity, and rainfall—can help improve WQI predictions. We used water and weather data from Cambridgeshire and Bedfordshire (UK) covering the years 2015 to 2024 and trained machine learning models, including Random Forest and Artificial Neural Networks, to predict WQI. The COVID-19 pandemic was treated as a natural event that allowed us to study water quality during reduced human activity. While weather data did not significantly boost model accuracy, it did help explain seasonal patterns and environmental shifts. Our analysis shows that water quality slightly improved during the COVID-19 lockdown years, followed by varying trends afterward. This research highlights the importance of combining water and climate data to better understand environmental changes and support future planning.**

***Index Terms*—Water Quality Index, COVID-19, Environmental Monitoring, Machine Learning, Artificial Neural Network, Random Forest, Weather Data Integration**

I. Introduction

Access to clean and safe water is essential for public health, economic development, and environmental sustainability. Nearly one in three people around the world lacks access to safe drinking water, making water quality monitoring a global priority (World Health Organization, 2023).

The Water Quality Index (WQI) is a well-established tool that combines several chemical and physical parameters—such as pH, dissolved oxygen, and nitrate—to represent the overall state of a water body. According to Tyagi, Sharma, Singh, and Dobhal (2013), WQI provides an effective means to evaluate and compare the quality of water across different locations and times by incorporating multiple physico-chemical parameters. However, most WQI prediction models rely solely on internal water chemistry and overlook the potential effects of external factors like weather. Rainfall, temperature, and humidity can significantly influence water quality by altering pollutant loads, runoff, and biological processes (U.S. Environmental Protection Agency, 2023; Yang et al., 2024).

The COVID-19 pandemic, which led to global reductions in industrial and transportation activity, created a unique opportunity to study how reduced human influence impacts environmental systems. Jawad-Ul-Haque, Arif, and Islam (2023) observed improvements in river pollution during lockdown periods in Bangladesh, while global reviews highlighted enhanced environmental quality across multiple domains (Loh, Tan, & Lee, 2022). However, only few studies have leveraged this event to analyze long-term WQI trends or to integrate weather data into predictive models.

This study aims to address two central research questions: First, can the inclusion of weather parameters—temperature, humidity, and precipitation—improve the predictive accuracy of machine learning models for WQI? Second, how did water quality change across the pre-COVID (2015–2019), during-COVID (2020–2021), and post-COVID (2022–2024) phases in the Cambridgeshire and Bedfordshire regions of the United Kingdom? By investigating these questions, the study seeks to develop a more comprehensive and context-aware approach to water quality prediction.

II. Review of Literature

Machine learning has shown significant promise in the field of environmental modeling, including water quality prediction. Prior research has demonstrated the robustness of ANN models in predicting water quality parameters in riverine environments (Singh, Basant, Malik, & Jain, 2009), which supports their selection in our study. Ehteram, Ahmed, Sherif, and El-Shafie (2024) demonstrated that deep learning models, such as Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs), can effectively estimate WQI using internal parameters like pH, nitrate, and dissolved oxygen. Their results achieved high accuracy, although their models did not

incorporate any external environmental variables. Similarly, explainable AI techniques have gained attention for improving transparency in model interpretation. Nallakaruppan et al. (2024) applied methods like SHAP and LIME to understand feature influence on WQI predictions, yet their scope was also limited to internal chemical metrics.

Environmental researchers have long recognized that weather plays a role in shaping water quality. Seasonal patterns of temperature and rainfall can influence pollutant concentration, microbial activity, and chemical interactions in water bodies (Singh et al., 2009). For example, recent work by Lin et al. (2024) demonstrated how different rainfall patterns significantly influence pollutant dispersion in river systems, highlighting the importance of integrating hydrological behavior in water quality assessments. Despite this, there are only few predictive WQI models incorporate such weather-related data into their feature sets.

The COVID-19 pandemic further highlighted the relationship between human activity and environmental health. In Bangladesh, significant reductions in heavy metal pollution were observed during the lockdown period (Jawad-Ul-Haque et al., 2023). Loh et al. (2022) reviewed global patterns and found widespread improvements in both air and water quality during the same period. While these studies provide valuable insights, they often rely on descriptive or comparative analyses rather than predictive modeling.

Despite the growing body of research in both machine learning and environmental monitoring, several critical gaps remain. Most machine learning models for WQI prediction ignore external factors such as climate, and very few studies extend across a full decade. Additionally, there is a lack of research that combines predictive modeling with analysis of societal disruptions like COVID-19.

This study addresses those gaps by integrating long-term water quality data with weather variables and evaluating machine learning performance across three pandemic-related phases. In doing so, it contributes to the broader goal of developing more adaptive and holistic frameworks for environmental monitoring.

III. Objectives of the Study

The primary objective of this study is to improve the prediction and understanding of water quality by integrating traditional water parameters with external environmental variables. While Water Quality Index (WQI) modeling has commonly relied on core chemical indicators such as pH, nitrate, and dissolved oxygen, this research explores the potential value of including weather-related data—specifically temperature, humidity, and precipitation—in machine learning models.

In addition to enhancing predictive capabilities, the study aims to examine how water quality has changed across three key timeframes defined by the COVID-19 pandemic: the pre-COVID period (2015–2019), the during-COVID period (2020–2021), and the post-COVID period (2022–2024). The pandemic serves as a natural experiment to explore how large-scale reductions in human activity influence environmental systems.

More specifically, the objectives of this study are as follows:

1) To predict WQI using machine learning models trained on both water quality and weather features.
2) To assess whether the inclusion of weather variables leads to measurable improvements in WQI prediction accuracy.
3) To analyze trends in WQI across three pandemic-related phases—pre-COVID, during-COVID, and post-COVID—within the Cambridgeshire and Bedfordshire regions.
4) To evaluate long-term water quality variations over a ten-year period (2015–2024), providing a broader context for understanding environmental shifts.
5) To explore the influence of the COVID-19 pandemic on water quality, using WQI as a proxy for ecological recovery during reduced anthropogenic stress.

IV. Data Collection

This study utilizes two primary datasets: water quality measurements from the Cambridgeshire and Bedfordshire regions of the United Kingdom, and historical weather records from nearby meteorological stations. Both datasets span the years 2015 to 2024.

Water quality data were obtained from the UK Environment Agency's Water Quality Archive (UK Environment Agency, 2024), an open-access platform offering monitored values for various rivers and water bodies. Key parameters included pH, dissolved oxygen, ammoniacal nitrogen, nitrate, orthophosphate, and water temperature. The data were initially recorded in a long format, with each row representing a single measurement per sampling event. This was later transformed into a structured wide format for analysis.

Weather data were retrieved from the Open-Meteo Historical Weather API (Open-Meteo, 2024), covering temperature, humidity, and precipitation. Records were collected from the Bedford and Cambridge weather stations and matched to the dates and locations of water samples.

After preprocessing, integration, and WQI calculation, the final dataset included 15,738 observations with 34 columns. These columns covered both water quality indicators and weather parameters. Key water features included pH, dissolved oxygen, nitrate, ammoniacal nitrogen, orthophosphate, and water temperature. Weather variables comprised daily maximum, minimum, and mean temperature; humidity; and precipitation. Geographic information (latitude, longitude) and temporal tags (date, year, COVID-phase) were also retained for spatial and trend-based analysis. The target variables for modeling were the computed WQI score (`WQI_Core_Final`).

To prepare the datasets for modeling, basic integration steps were performed. These included aligning date formats, standardizing measurement units, and mapping water samples to the nearest weather station using spatial coordinates. Further details on preprocessing are discussed in the methodology section.

A focused subset of variables was selected for further analysis and model development. These included six core water quality indicators and three key weather parameters believed to influence water conditions. The computed Water Quality Index (`WQI_Core_Final`) is used as the target variable. Table I summarizes the selected variables, their measurement units, and brief descriptions of their roles in the study.

V. Methodology

As shown in Figure 1, the process begins with data acquisition, followed by parameter selection, imputation, and dataset merging. Water Quality Index (WQI) is then calculated using selected parameters. The merged dataset is used both for regression-based prediction modeling and for COVID-phase-wise trend analysis. Results are then visualized, interpreted, and used to derive final conclusions.

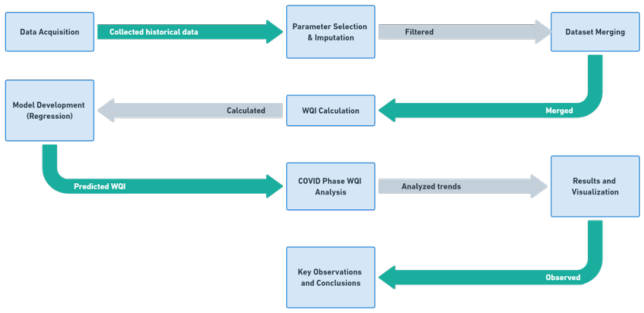

Fig. 1. Overview of the complete workflow used in this study.

A. Exploratory Data Analysis (EDA)

Exploratory data analysis (EDA) was conducted to better understand the characteristics of the dataset and to prepare it for modeling. The dataset spanned the years 2015 to 2024 and included core water quality parameters, selected weather variables, and calculated Water Quality Index (WQI) values. Special focus was placed on handling missing data, identifying outliers, and labeling samples across COVID-related phases to facilitate comparative analysis.

As shown in Figure 3, average WQI scores increased slightly during the pandemic years. However, in Figure 2, the number of water quality samples dropped significantly in 2020 and 2021, likely due to restricted field operations and halted monitoring activities during COVID-19 lockdowns. This reduced sampling density may affect the robustness and generalizability of phase-based comparisons.

B. Hypotheses for the Study

Two guiding hypotheses were developed for the analytical phase:

Hypothesis 1: Impact of Weather Variables on WQI Prediction

- $\mathbf{H_1}$: Integrating weather variables (temperature, humidity, precipitation) into machine learning models significantly improves WQI prediction accuracy.

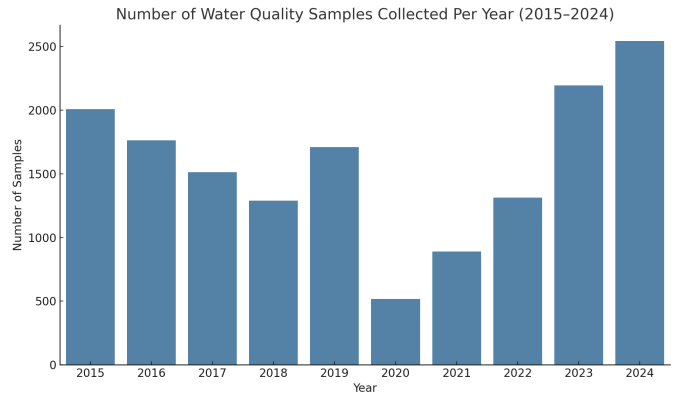

Fig. 2. Annual count of water quality samples collected from 2015 to 2024.

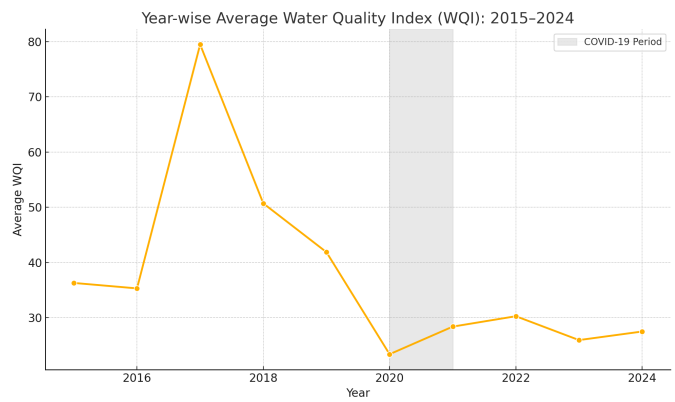

Fig. 3. Year-wise average Water Quality Index (WQI) from 2015 to 2024, based on samples from Cambridgeshire and Bedfordshire. The shaded region (2020–2021) marks the COVID-19 lockdown period, which coincided with a slight improvement in WQI.

- $\mathbf{H_0}$: Integrating weather variables does not significantly improve WQI prediction accuracy compared to using water parameters alone.

Hypothesis 2: Impact of COVID-19 on Water Quality

- $\mathbf{H_1}$: Water Quality Index (WQI) improved during the COVID-19 lockdown period due to reduced human activity.
- $\mathbf{H_0}$: There is no significant difference in WQI levels between pre-COVID, during-COVID, and post-COVID periods.

C. Preprocessing

1) Missing Value Analysis: The original water quality dataset exhibited substantial missingness, especially in less frequently monitored parameters. Parameters with more than 50% missing values were excluded from the analysis to maintain consistency and avoid bias introduced by excessive imputation. For the remaining parameters, time-based interpolation was applied within each sampling point to estimate missing values where feasible. Records with unresolved missing values in critical features were dropped to ensure the overall integrity of the dataset.

TABLE I
Key Features Used in WQI Calculation and Prediction

Variable	Unit	Description
pH	ph units	Acidity/alkalinity level of the water
Ammoniacal Nitrogen as N	mg/L	Concentration of ammoniacal nitrogen
Nitrate as N	mg/L	Concentration of nitrate
Orthophosphate, reactive as P	mg/L	Concentration of reactive orthophosphate
Oxygen, Dissolved as O_2	mg/L	Dissolved oxygen content in the water
Temperature of Water	°C	Measured temperature of the water sample
WQI_Core_Final	Index value	Final computed Water Quality Index (core version)
Temperature (air, mean)	°C	Daily average air temperature from nearest station
Relative Humidity (mean)	%	Daily average relative humidity
Precipitation (sum)	mm	Daily total precipitation (rainfall)

2) Outlier Detection and Treatment: Outliers were identified using the Interquartile Range (IQR) method, focusing on highly variable parameters such as nitrate and ammoniacal nitrogen. These values were often linked to localized spikes in pollution or sampling anomalies. Rather than removing these records, a smoothing strategy was applied by replacing extreme values with the average of their nearest non-outlier neighbors. This approach helped retain overall sample volume while minimizing distortion during model training.

3) COVID Phase Tagging: Each sample was assigned to one of three COVID-related temporal phases based on its timestamp:

- **Pre-COVID:** 2015–2019
- **During COVID:** 2020–2021
- **Post-COVID:** 2022–2024

This tagging allowed phase-wise trend analysis and supported one of the study's key hypotheses regarding environmental changes during the pandemic.

D. Water Quality Index (WQI) Calculation

The WQI was computed using the Weighted Arithmetic Index Method, a widely accepted approach for water quality assessment (Ehteram et al., 2024; Lukhabi, Mensah, Asare, Pulumuka-Kamanga, & Ouma, 2023). This method follows the approach outlined by Tyagi et al. (2013), which assigns weights to each parameter based on its relative importance and calculates a composite index to classify the water quality.

Weights (W_i) were calculated based on the inverse of each parameter's standard permissible value (S_i):

$$W_i = \frac{1/S_i}{\sum_{j=1}^{n}(1/S_j)} \tag{1}$$

This ensures that all weights sum to 1 and that parameters with stricter permissible limits contribute more significantly to the overall WQI score. For example, ammoniacal nitrogen and orthophosphate, which are harmful even at low concentrations, were given higher relative weights compared to less sensitive parameters like nitrate.

The quality rating for each parameter (Q_i) was calculated as:

$$Q_i = \frac{(V_i - V_0)}{(S_i - V_0)} \times 100 \tag{2}$$

The final WQI was then computed using:

$$\text{WQI} = \frac{\sum(W_i \times Q_i)}{\sum W_i} \tag{3}$$

Where:

- V_i is the observed concentration of parameter i
- V_0 is the ideal value for that parameter
- S_i is the standard permissible value
- W_i is the weight for parameter i, inversely proportional to S_i

1) WQI Parameters and Standards: The parameters used in the WQI calculation and their respective ideal (V_0) and permissible (S_i) values are shown in Table II. These were adopted from standard environmental benchmarks and prior research (Ehteram et al., 2024; Lukhabi et al., 2023).

TABLE II
Standard and Ideal Values of Selected Water Quality Parameters for WQI Calculation

Parameter	Vi (Ideal)	Si (Standard Permissible)
pH	7.0	8.5
Ammoniacal Nitrogen (mg/L)	0.0	0.5
Orthophosphate (mg/L)	0.0	0.6
Temperature of Water (°C)	0.0	25.0
Dissolved Oxygen (mg/L)	14.0	5.0
Nitrate (mg/L)	0.0	45.0

E. Feature Selection for Modeling

Based on data availability and domain knowledge, a focused subset of variables was selected for modeling. This included six water quality parameters (pH, dissolved oxygen, nitrate, ammoniacal nitrogen, orthophosphate, and water temperature) and three weather parameters (temperature, humidity, and precipitation). The computed WQI (`WQI_Core_Final`) served as the target variable, with its corresponding categorical label (`WQI_Class`) used for phase-based comparisons.

F. Model Development and Evaluation

To predict the Water Quality Index (WQI), two regression-based models were developed: a Random Forest Regressor and an Artificial Neural Network (ANN). These models were chosen to compare ensemble-based learning and deep learning in handling environmental data with both structured inputs and moderate feature interactions.

1) Random Forest Regressor: The Random Forest model was implemented using Scikit-learn's (`RandomForestRegressor`). It comprised an ensemble of 100 decision trees, each trained on random subsets of the data to reduce variance and overfitting. The model is known for its robustness to outliers and its ability to handle non-linear relationships among features. Hyperparameters such as the number of estimators (`n_estimators=100`) were selected based on standard practice, and the model was trained on both scaled and unscaled feature sets, depending on the configuration (water-only vs. combined with weather data).

2) Artificial Neural Network (ANN): ANN has been successfully used in several previous studies to predict water quality, thanks to its ability to capture nonlinear relationships in environmental data (Singh et al., 2009). The ANN model in this study was implemented using the `MLPRegressor` class from Scikit-learn. The architecture consisted of:

- Two hidden layers with 64 and 32 neurons respectively
- ReLU activation function for non-linearity
- Adam optimizer for efficient stochastic gradient-based learning
- Maximum iterations set to 1000 with automatic early stopping

This configuration was chosen for its balance between model complexity and interpretability. Unlike the base paper, which used deeper architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), this simpler feedforward network was more suitable for our dataset's size and the need for explainable results. The ANN captured complex interactions among water and weather features while remaining generalizable through cross-validation.

3) Cross-Validation and Evaluation Strategy: To ensure that model performance was not biased by a specific train-test split, 5-Fold Cross-Validation was applied, particularly for the Artificial Neural Network (ANN) model. The data was randomly partitioned into five folds; in each iteration, one fold was used as the validation set while the remaining four were used for training.

This method helps assess the model's generalizability and stability across different subsets of the data, which is especially important given the natural variability in environmental datasets. The cross-validation scores were aggregated using the R^2 metric to evaluate how well the model could explain the variance in WQI across unseen data.

For all models, the following evaluation metrics were used:

- Mean Absolute Error (MAE): Average magnitude of absolute prediction errors
- Root Mean Squared Error (RMSE): Penalizes larger errors more heavily, giving a sense of prediction reliability
- R^2 Score (Coefficient of Determination): Proportion of variance in WQI explained by the model

These metrics were calculated separately for:

- Models trained using only water quality parameters
- Models trained using combined water and weather parameters

VI. Data Visualization

This section presents the key findings of the study through visual analysis and model evaluation. The results are organized to highlight temporal trends in WQI, the role of weather variables, the impact of COVID-19 phases, and the performance of predictive models.

A. Average WQI vs Rainfall Trends (2015–2024)

A dual-axis time series plot (Figure 4) compares the average Water Quality Index (WQI) to average rainfall between 2015 and 2024. Although no consistent linear relationship is observed between rainfall and WQI, a notable shift in the pattern emerges around the COVID-19 period (2020–2021). During these years, WQI values reached lower levels, coinciding with reduced industrial and agricultural activities. Post-pandemic, both rainfall and WQI gradually recovered, though with some fluctuations. This shift may indicate a temporary environmental reset in response to decreased anthropogenic pressure.

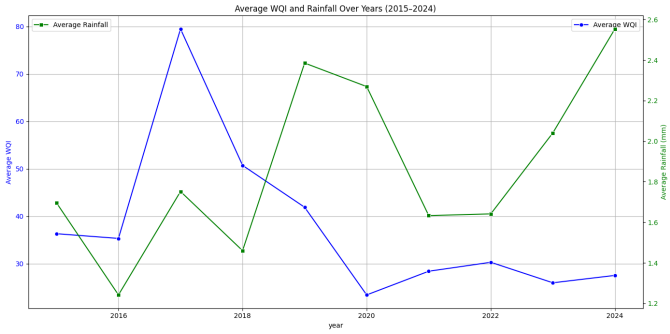

Fig. 4. Average WQI and Rainfall Over Years

B. Average WQI vs Water Temperature Trends

In Figure 5, average water temperature is compared with WQI values across the study period. While direct correlation is not strong year-to-year, the COVID period again marks a clear deviation from earlier trends — with WQI dropping and

temperature reaching a local minimum. This inverse behavior suggests that environmental conditions may have collectively shifted, potentially due to lower surface water contamination and altered climate interactions during lockdowns. The lack of a consistent pattern across the decade suggests that temperature alone is not a primary driver of WQI, but it may contribute contextually during certain events such as the pandemic.

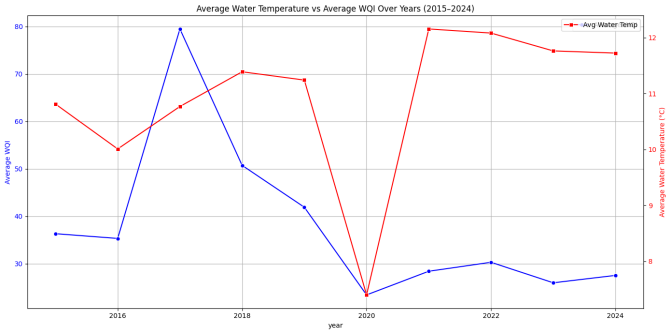

Fig. 5. Average Water Temperature vs Average WQI Over Years

C. WQI Class Distribution Across COVID Phases

Figure 6 displays the distribution of WQI classes across the Pre-COVID (2015–2019), During-COVID (2020–2021), and Post-COVID (2022–2024) phases. The chart shows a clear increase in "Excellent" and "Good" classifications during the COVID period, likely due to reduced human activity. After the pandemic, a moderate decline in water quality is observed, evidenced by the resurgence of "Poor" and "Very Poor" categories.

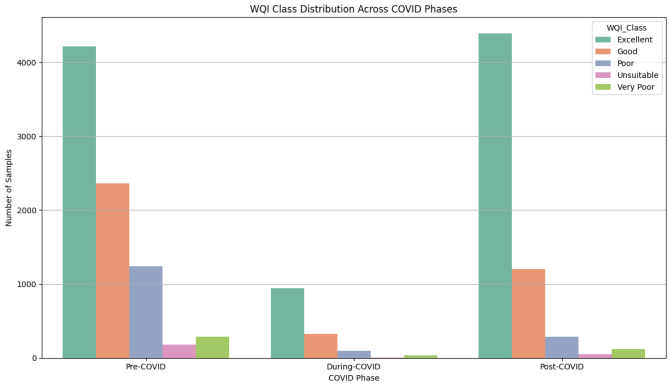

Fig. 6. WQI Class Distribution Across COVID Phases

D. Pollutant Concentration Trends Across Phases

Figure 7 shows average concentrations of four key parameters—Ammoniacal Nitrogen, Orthophosphate, Dissolved Oxygen, and Nitrate—across the COVID phases. Results indicate:

- Ammoniacal Nitrogen and Orthophosphate levels decreased significantly during COVID and rose again post-pandemic.
- Dissolved Oxygen peaked during COVID, reflecting reduced organic pollution.
- Nitrate remained relatively stable, suggesting limited influence from transient surface events.

These observations explain the quantitative improvement in WQI during lockdown years, driven primarily by reductions in nutrient pollutants.

VII. Predictive Model Results

A. Model Comparison Summary

These results from Table III show that the Artificial Neural Network (ANN) significantly outperformed the Random Forest model, especially when trained on water-only features. Surprisingly, the addition of weather data did not improve model accuracy and slightly reduced performance in the ANN model. This finding aligns with earlier visualizations suggesting that weather parameters have weak direct influence on WQI in this region.

TABLE III
Final Model Comparison Summary

Model	MAE	RMSE	R^2
Random Forest (Water-Only)	1.942	51.74	0.9470
Random Forest (Water + Weather)	1.892	43.56	0.9624
ANN (Water-Only)	0.283	1.356	0.9999
ANN (Water + Weather)	0.483	1.879	0.9999

B. Cross-Validation of ANN

From Figure 8, using 5-Fold Cross-Validation on the ANN model, the average R^2 score achieved was 0.9956, indicating high generalizability across different data subsets and confirming that the model is not overfitting to the training data.

C. Residual Analysis (ANN Model)

Residual analysis (Figure 9) reveals a random scatter of residuals around zero, with no visible pattern, confirming that the ANN model's predictions are well-balanced and free from systemic bias.

D. Hypothesis Validation

The findings of this study offer direct insights into the hypotheses proposed earlier.

Hypothesis 1 — that integrating weather parameters would improve WQI prediction — is **not supported**. While weather data was incorporated into both Random Forest and ANN models, its inclusion did not enhance model performance and in some cases slightly reduced it, especially in the ANN model. This suggests that weather variables such as rainfall, temperature, and humidity may not significantly influence WQI in the studied region, at least in a predictive modeling context.

In contrast, **Hypothesis 2** — that water quality improved during the COVID-19 lockdown — is **strongly supported**. Visual analyses showed higher proportions of "Excellent" and "Good" WQI classifications during the pandemic period. Additionally, pollutant-level comparisons confirmed measurable decreases in ammoniacal nitrogen and orthophosphate

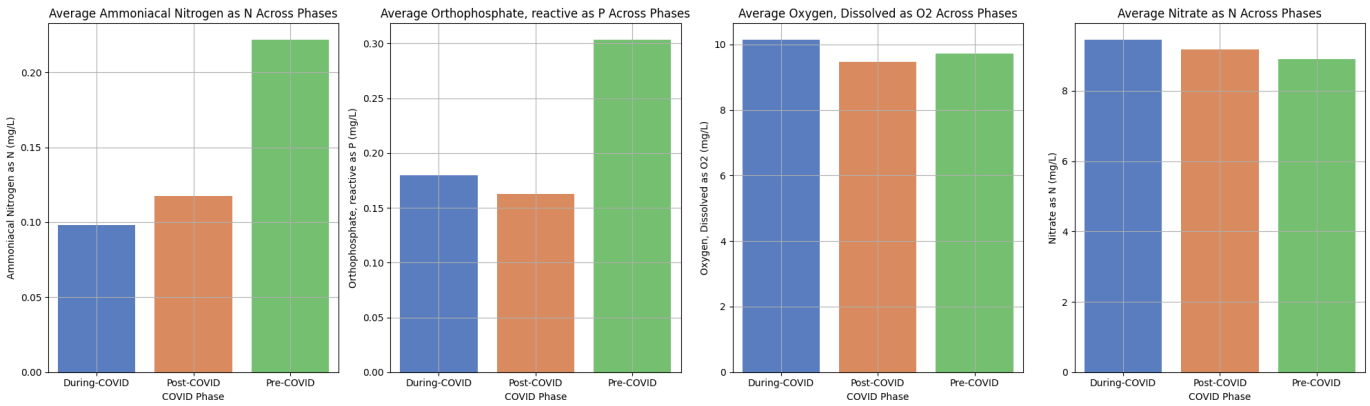

Fig. 7. Key Pollutants Across COVID Phases

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Cross-Validation R² Scores: [0.99996712 0.99994053 0.97846807 0.99974011 0.99993357]
Average CV R²: 0.9956
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Fig. 8. Cross-Validation K-Fold

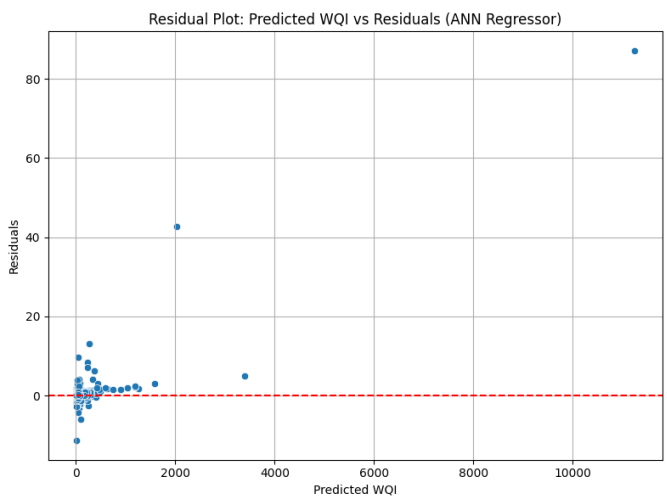

Fig. 9. Residual Plot – Predicted WQI vs Residuals

concentrations, accompanied by increased dissolved oxygen levels. These findings align with the hypothesis that reduced human activity during the lockdown contributed to short-term environmental recovery.

Although no formal statistical hypothesis testing (e.g., ANOVA) was conducted, the hypotheses were evaluated through visual comparisons of WQI trends, phase-based summaries, and model performance metrics.

VIII. Conclusion

This study explored the prediction and analysis of Water Quality Index (WQI) across the Cambridgeshire and Bedfordshire regions from 2015 to 2024, integrating both water quality and weather datasets. The primary goals were to assess the impact of weather variables on WQI prediction and to evaluate water quality trends across pre-COVID, during-COVID, and post-COVID phases.

Using a systematic pipeline involving data cleaning, WQI calculation, and machine learning modeling, two predictive models—Random Forest and Artificial Neural Network (ANN)—were trained and evaluated. The ANN model, particularly when using only water quality parameters, achieved outstanding performance ($R^2 > 0.999$), far outperforming Random Forest and confirming the strength of neural networks in capturing environmental patterns. Surprisingly, the inclusion of weather features such as temperature, humidity, and precipitation had little to no impact on model accuracy, indicating that WQI is primarily driven by intrinsic water chemistry in this region.

The analysis of pollutant concentrations and WQI class distributions during the COVID-19 lockdown years revealed a temporary improvement in water quality, driven by reductions in ammoniacal nitrogen and orthophosphate levels. These findings support the hypothesis that reduced human activity during the pandemic contributed to short-term environmental recovery. Meanwhile, the lack of improvement from weather data integration challenges the assumption that climate variables significantly influence WQI in this context.

Like any research, this study has its limitations. The dataset lacked certain pollutant indicators (e.g., biological contaminants, turbidity, heavy metals), and sampling density varied year to year, especially during COVID-19. Additionally, weather features were limited to surface-level metrics without accounting for broader hydrological or land-use dynamics. Future research can build on this work by incorporating a wider range of environmental variables, exploring spatial modeling approaches, validating findings across different regions, and developing real-time prediction systems using high-frequency or satellite data.

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